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```
import time
def merge_sort(arr):
    if len(arr) > 1:

        mid = len(arr) // 2
        left = arr[:mid]
        right = arr[mid:]

        merge_sort(left)
        merge_sort(right)

        i = j = k = 0

        while i < len(left) and j < len(right):
            if left[i][0] <= right[j][0]:
                arr[k] = left[i]
                i += 1
            else:
                arr[k] = right[j]
                j += 1
            k += 1

        while i < len(left):
            arr[k] = left[i]
            i += 1
            k += 1

        while j < len(right):
            arr[k] = right[j]
            j += 1
            k += 1

data = [(8, '8'), (3, 'A'), (5, '5A'),
        (3, 'B'), (9, '9'),
        (1, '1'), (5, '5B'), (2, '2')]

start = time.perf_counter()

merge_sort(data)

end = time.perf_counter()

print("Sorted Array")
print(data)

print("Execution Time =", end-start)
```

```
Python 3.14.4 (tags/v3.14.4:23116f9, Apr 7 2026, 14:10:54) [MSC v.1944 64 bit (AMD64)] on win32
Enter "help" below or click "Help" above for more information.
>>> = RESTART: C:/Users/jasmi/OneDrive/Pictures/Documents/daa python code asses.py =
Sorted Array
[(1, '1'), (2, '2'), (3, 'A'), (3, 'B'), (5, '5A'), (5, '5B'), (8, '8'), (9, '9')]
Execution Time = 1.1499971151351929e-05
>>> |
```

Ln: 8 Col: 0